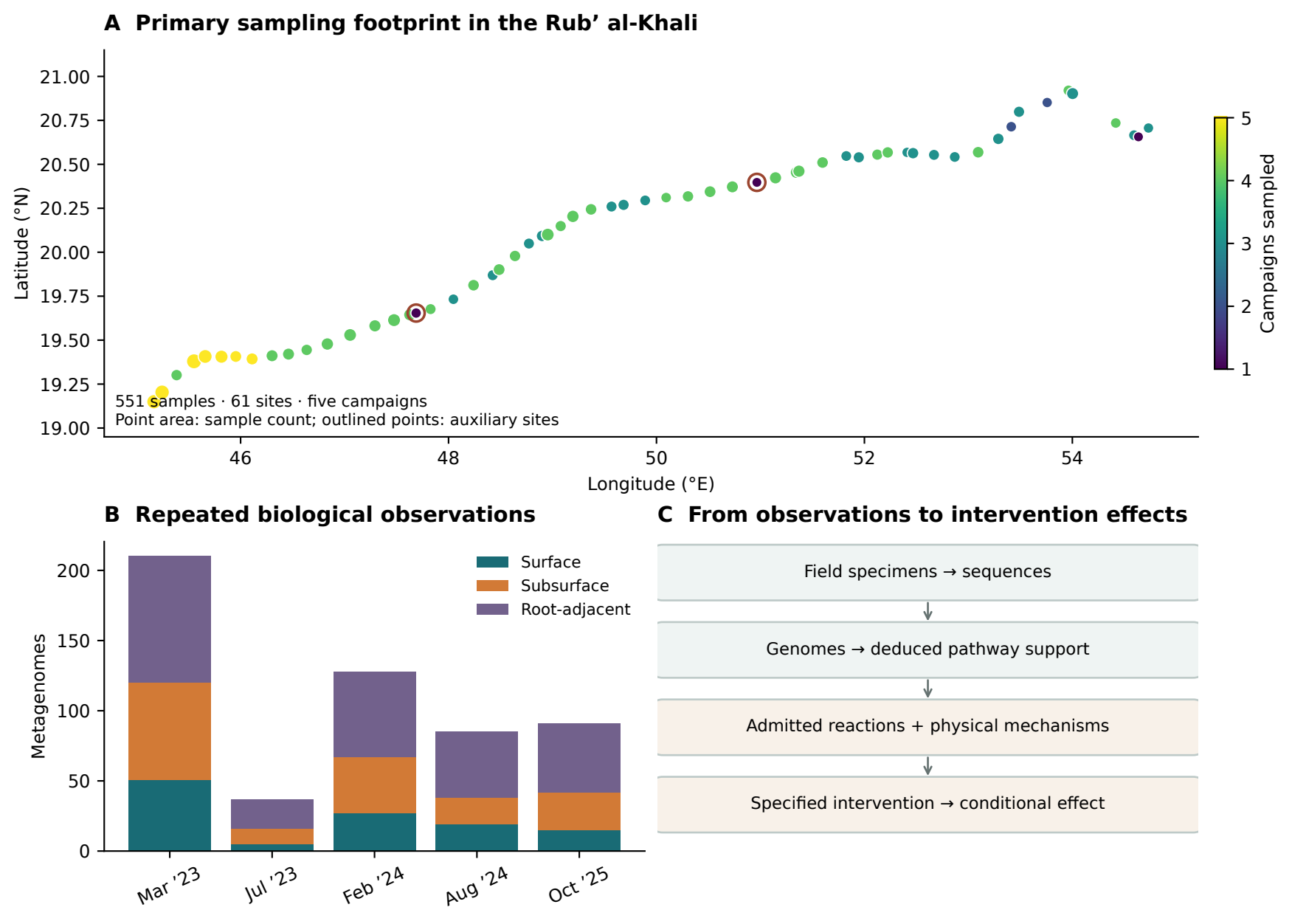
A Primary sampling footprint in the Rub’ al-Khali
Latitude (°N)
21.00
20.75
20.50
20.25
20.00
19.75
19.50
19.25
19.00
46
48
50
52
54
Longitude (°E)
551 samples · 61 sites · five campaigns
Point area: sample count; outlined points: auxiliary sites
Campaigns sampled
5
4
3
2
1
B Repeated biological observations
Metagenomes
200
150
100
50
0
Surface
Subsurface
Root-adjacent
Mar '23
Jul '23
Feb '24
Aug '24
Oct '25
C From observations to intervention effects
Field specimens → sequences
Genomes → deduced pathway support
Admitted reactions + physical mechanisms
Specified intervention → conditional effect